Unified Salt–Ice–Water Model, with the Can

Full equation compendium and step-by-step derivations: second edition

Styrofoam heat transfer, ice melting, NaCl dissolution, and a drink can that may be part-filled and may contain ethanol

SYPT modelling notes

29 October 2026

Abstract

This second edition expands every step of the first (26 October 2026) and adds the can of drink that the bath is there to cool. Each equation is derived from the ones before it, line by line, with the assumption it needs, a check, and a worked number. Steps 1–7 are the bath, as before: the first law, the heat that leaks in through the Styrofoam, the ice, the water, the heat of dissolving salt, the freezing point and the rate at which salt dissolves. Steps 8–10 are new: the drink and its can as a second body that gives heat to the bath; a can that is not completely full, standing or lying, held under or floating; and an ethanol model of the drink, which fixes its heat capacity and how cold it can get before it starts to turn to slush. Steps 11 and 12 rebuild the time-dependent and equilibrium models with the can in them. As in the first edition, equations that need qualification or correction are labelled; they should not be used as exact identities.

Contents

What is new in this edition, and how to read it **3**

1 System, notation and assumptions **3**

2 Step 1: energy in = energy out **5**

3 Step 2: heat entering through Styrofoam **6**

4 Step 3: the ice and the latent heat **7**

5 Step 4: liquid water and mass conservation **8**

6 Step 5: enthalpy of dissolution of NaCl **8**

7 Step 6: freezing-point depression $i_v K_f b$ **9**

8 Step 7: salt dissolution rate and shrinking crystals **11**

9 Step 8: the can and its drink (new) **12**

9.1 The drink as one body 12

9.2 The heat path through the wall 12

9.3 The film coefficients . . . 12
9.4 Is one drink temperature enough? . . . 13
9.5 The fixed-bath solution . . . 13

10 Step 9: a can that is not completely full (new) 13
10.1 What the headspace changes . . . 13
10.2 Measuring the fill . . . 14
10.3 Upright can . . . 14
10.4 Lying can: the circular segment . . . 14
10.5 Does the can float? . . . 14
10.6 What the fill and the immersion do to the cooling . . . 15
10.7 Room to freeze . . . 16

11 Step 10: the drink as a solution, an ethanol model (new) 16
11.1 Composition from the label . . . 16
11.2 Where the drink starts to freeze . . . 17
11.3 Freezing gradually: slush . . . 17
11.4 The drink's heat capacity . . . 18
11.5 Convection inside the can . . . 18
11.6 Supercooling . . . 18
11.7 What ethanol changes . . . 18
11.8 Ethanol in the bath instead of salt . . . 18

12 Step 11: fully unified time-dependent model, with the can 19

13 Step 12: time-independent equilibrium, with the can 21
13.1 Insulated equilibrium . . . 21
13.2 Integrated heat leakage over a finite interval . . . 22
13.3 True steady state with heat leakage . . . 23

14 A thermodynamically more consistent refinement 23
14.1 Consistent integrated energy at any starting and final temperature . . . 23

15 Model limits and practical next steps 24

16 Quick index of principal equations 24

What is new in this edition, and how to read it

The first edition modelled the bath alone: ice, brine and salt in a Styrofoam box. A bath is only there to cool something, so this edition adds the can and its drink, and expands every step.

- **The can** (Steps 8–10). The drink and its can are a second body, with its own temperature T_d, that gives heat to the bath through the can wall. Step 8 derives that heat flow. Step 9 works out which part of the wall passes it when the can is not full, stands or lies, is held under or floats. Step 10 models the drink as a solution of ethanol, sugar and carbon dioxide in water, which fixes its heat capacity and the temperature at which it starts to freeze.
- **The coupled models** (Steps 11–12). The time-dependent model and the equilibrium of the first edition are rebuilt with the drink's heat in them, and solved for the rig.
- **Every step expanded.** Each step says what it does and what it assumes, derives each equation from the one before, checks it, and works a number.
- **New corrections**, labelled like those of the first edition: the Styrofoam's air films (Step 2); the salt's own heat capacity (Step 5); the error of the dilute freezing law in strong brine, and the temperature at which the model's bath stops cooling (Step 6); and starting temperatures other than 0 °C in the equilibrium (Step 12).

How each step is laid out. The idea says in words what the step does. **Assumptions** lists what it takes for granted. **Derivation** goes line by line: every equation follows from the ones before it, and the text says how. Key results are boxed. **Check** tests a result against its units or a limiting case. A shaded box works a number. **Where it fails** says when not to use the result.

Two sets of numbers. The first edition illustrated its equations with a small demonstration bath. A can needs a bath big enough to cool it, so the new steps use the rig of the B8 experiment with Can 2. Both sets are in Table 1; each worked example says which it uses.

Table 1: The two sets of numbers used in the worked examples.

	Demonstration (first edition)	Rig, with Can 2
Box	$k = 0.033\,\mathrm{W\,m^{-1}K^{-1}}$, $A = 0.05\,\mathrm{m^2}$, $d = 0.02\,\mathrm{m}$, so $H = 0.0825\,\mathrm{W\,K^{-1}}$	30 mm of expanded polystyrene: $H \approx 0.46\,\mathrm{W\,K^{-1}}$ for the foam alone
Room	25 °C	28 °C
Ice	0.200 kg at 0 °C	4.0 kg at 0 °C
Water	0.100 kg at 0 °C	1.6 kg at 4 °C
Salt	0.050 kg	1.0 kg at 28 °C, grains 0.5 mm across
Drink	—	330 g at 28 °C: a cola, or a 5 % beer (Step 10)
Can	—	aluminium, 13 g; diameter 6.6 cm, height 11.66 cm; headspace 1.2 cm when upright
Films	—	$h_i = 400$, $h_o = 270\,\mathrm{W\,m^{-2}K^{-1}}$ upright; both $\times 1.2$ lying (Step 8)

1 System, notation and assumptions

A closed Styrofoam box holds a container of ice, liquid water (brine, once salt dissolves), solid NaCl and dissolved NaCl: the *bath*. A sealed can of drink sits in the bath. The room, at T_{room}, leaks heat into the bath through the box. The drink, at T_d, gives heat to the bath through the can wall. The bath, at T, absorbs both by warming, by melting ice and by dissolving salt.

Let $m_i(t)$, $m_w(t)$ and $m_s(t)$ be the masses of ice, liquid water and *dissolved* NaCl (kg). Let $m_{s,0}$ be the solid salt added at the start and $m_{s,d,0}$ any salt dissolved at the start. Then $M_w = m_{i,0} + m_{w,0}$

is all the water and $S_0 = m_{s,0} + m_{s,d,0}$ all the salt. Temperatures in sensible-heat terms are in °C (a temperature difference is the same in K); absolute temperature appears only in Step 6, written $\mathcal{T}$.

Assumptions about the bath.

B1 One bath temperature. One temperature T describes the whole bath. With no stirrer this is an idealisation: brine next to the can warms first (the B8 report models that spreading).

B2 Ice and brine in equilibrium. Ice melts or freezes fast enough that, while both are present, the bath sits at the freezing point of its brine (Step 6).

B3 Salt dissolves at a finite rate (Step 7), so the dissolved salt is a variable of its own.

B4 Constant properties. Specific heats, the latent heat and the heat of solution are constants. The brine's heat capacity is that of its water.

B5 The box is a steady conductor (Step 2): it stores no heat and passes heat in proportion to the temperature difference.

B6 Closed and still. The lid is on, nothing evaporates, nothing is added during a run, and nothing does work on the bath.

Assumptions about the can and drink.

C1 One drink temperature T_d: free convection keeps the drink mixed (Step 8).

C2 Steady wall and films: the can's wall and the liquid films on it store no heat.

C3 A sealed can: the drink's mass is fixed, and the gas in the headspace holds negligible heat.

C4 Heat passes only where the wall has drink inside and brine outside (Step 9).

C5 The drink's properties follow from its composition (Step 10).

Table 2: Notation. Numbers are the constants used throughout; the rest is set by the experiment.

Symbol	Meaning	Value or unit
The bath (first edition)		
k	Styrofoam conductivity	$\mathrm{W\,m^{-1}K^{-1}}$
$A,\ d$	area and thickness of the insulating layer	$\mathrm{m^2}$, m
$H = kA/d$	conductance of the box	$\mathrm{W\,K^{-1}}$
$T,\ T_{\mathrm{room}}$	bath and room temperatures	°C
$m_i,\ m_w,\ m_s$	ice, liquid water, dissolved NaCl	kg
$M_w,\ S_0$	all the water, all the salt	kg
$W = M_w - m_i$	liquid water, equal to m_w	kg
$c_i,\ c_w$	specific heats of ice and liquid water	2100, $4180\,\mathrm{J\,kg^{-1}K^{-1}}$
c_s	specific heat of solid NaCl	$864\,\mathrm{J\,kg^{-1}K^{-1}}$
L_f	latent heat of melting ice at 0 °C	$333\,500\,\mathrm{J\,kg^{-1}}$
M_{NaCl}	molar mass of NaCl	$0.05844\,\mathrm{kg\,mol^{-1}}$
ΔH_{sol}	molar heat of solution of NaCl	$3900\,\mathrm{J\,mol^{-1}}$
$\beta = \Delta H_{\mathrm{sol}}/M_{\mathrm{NaCl}}$	heat of solution per kg	$66\,735\,\mathrm{J\,kg^{-1}}$
K_f	cryoscopic constant of water	$1.86\,\mathrm{K\,kg\,mol^{-1}}$
i_v	van 't Hoff factor: ions per formula unit (named i_v to avoid the ice index)	2
$\alpha = i_v K_f/M_{\mathrm{NaCl}}$	freezing-point coefficient	63.65 K
b	molality: moles of solute per kg of liquid water	$\mathrm{mol\,kg^{-1}}$
k_m	dissolution mass-transfer coefficient	$\mathrm{m\,s^{-1}}$
A_s	exposed surface area of the salt crystals	$\mathrm{m^2}$
$C,\ C_{\mathrm{sat}}$	salt per volume of water; the same at saturation	$\mathrm{kg\,m^{-3}}$; $C_{\mathrm{sat}} \approx 357$
ρ_w	density of water	$1000\,\mathrm{kg\,m^{-3}}$

Symbol	Meaning	Value or unit
$C_{\text{eff}} = m_i c_i + W c_w$	heat capacity of the ice and water	$\mathrm{J\,K^{-1}}$
The can and drink (new)		
T_d	drink temperature	°C
$m_d,\ c_d$	drink mass; drink specific heat (Step 10)	kg; $\mathrm{J\,kg^{-1}K^{-1}}$
$m_c,\ c_c$	can mass; specific heat of aluminium	kg; $900\,\mathrm{J\,kg^{-1}K^{-1}}$
$C_d = m_d c_d + m_c c_c$	heat capacity of the drink and can	$\mathrm{J\,K^{-1}}$
$r,\ \ell$	radius and height of the can	m
$t_w,\ k_w$	wall thickness; conductivity of can aluminium	m; $160\,\mathrm{W\,m^{-1}K^{-1}}$
$h_i,\ h_o$	drink-side and brine-side film coefficients	$\mathrm{W\,m^{-2}K^{-1}}$
h_{ov}	overall coefficient through both films and the wall	$\mathrm{W\,m^{-2}K^{-1}}$
A_c	area of wall with drink inside and brine outside	$\mathrm{m^2}$
$G = h_{\text{ov}} A_c$	conductance from the drink to the bath	$\mathrm{W\,K^{-1}}$
$\dot{Q}_d = G(T_d - T)$	heat flow from the drink to the bath	W
h_g	headspace height, can upright	m
f	fill fraction: share of the can's volume that is drink	—
$z_d,\ z_b$	upright can: drink height; depth of brine on the can	m
$y_d,\ y_g,\ y_b$	lying can: drink depth; gas-strip depth; depth of brine on the can	m
$A_{\text{seg}}(y),\ P(y)$	area and arc length of a circular segment of depth y	$\mathrm{m^2}$, m
$\rho_d,\ \rho_b$	densities of the drink and of strong brine	$\mathrm{kg\,m^{-3}}$; $\rho_b \approx 1182$
The drink as a solution (new)		
ABV	alcohol by volume	—
$w_e,\ w_{su},\ w_c,\ w_x$	mass fractions of ethanol, sugar, CO_2 and other extract in the drink	—
$w_{wd},\ m_{wd}$	mass fraction and mass of water in the drink	—, kg
$M_e,\ M_{su},\ M_c$	molar masses: ethanol, sucrose, CO_2	0.04607, 0.3423, $0.04401\,\mathrm{kg\,mol^{-1}}$
ρ_e	density of ethanol at 20 °C	$789.2\,\mathrm{kg\,m^{-3}}$
T_{fd}	temperature at which the drink starts to freeze	°C
ψ	share of the drink's water that is frozen	—
$C_{d,\text{eff}}$	effective heat capacity of a freezing drink	$\mathrm{J\,K^{-1}}$

2 Step 1: energy in = energy out

The idea. Energy is not created or destroyed. Heat that enters the bath either stays in it, as a change of its energy, or leaves again. The can adds a second body to keep account of.

The first edition's framing.

$$E_{\text{in}} = E_{\text{out}}, \tag{1}$$

$$E_{\text{in}} - E_{\text{out}} = \Delta U, \qquad U_{\text{initial}} = U_{\text{final}} \ \text{ if insulated and no work.} \tag{2}$$

Derivation. The first law of thermodynamics for a *closed* system, one that no mass crosses, says that its internal energy U changes by the heat Q that enters it minus the work W_{by} it does on its surroundings: $\Delta U = Q - W_{\text{by}}$. Writing the net heat as what comes in minus what goes out, $Q = E_{\text{in}} - E_{\text{out}}$, and taking $W_{\text{by}} = 0$ gives (2).

Why the work is zero: the bath does no work except expanding against the atmosphere, and for liquids and solids that is tiny. Melting 1 kg of ice shrinks it by $1/917 - 1/1000 = 9.1 \times 10^{-5}\,\mathrm{m^3}$; at atmospheric pressure that is $p\,\Delta V \approx 9\,\mathrm{J}$, against the 333 500 J the melting takes. So here the

internal energy U and the enthalpy $H = U + pV$ change by the same amount, and either may be used.

(1) is the special case $\Delta U = 0$. It holds only when nothing accumulates, for example in a steady state. While the bath cools or melts, its energy changes, and (2) is the correct statement. Dividing by a short time Δt and letting it shrink turns energies into rates:

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \dot{Q}_{\mathrm{in}}, \tag{3}$$

with $\dot{Q}_{\mathrm{in}}$ the net heat flow into the system, in watts.

Two bodies (new). With the can there are two closed systems. The bath B receives heat through the box, $\dot{Q}_{\mathrm{box}}$, and from the drink, $\dot{Q}_d$; the drink and can D lose $\dot{Q}_d$. Applying (3) to each, and adding:

$$\frac{\mathrm{d}U_B}{\mathrm{d}t} = \dot{Q}_{\mathrm{box}} + \dot{Q}_d, \tag{4}$$

$$\frac{\mathrm{d}U_D}{\mathrm{d}t} = -\dot{Q}_d, \tag{5}$$

$$\frac{\mathrm{d}(U_B + U_D)}{\mathrm{d}t} = \dot{Q}_{\mathrm{box}}. \tag{6}$$

$\dot{Q}_d$ is counted positive from the drink to the bath, so it leaves one body and enters the other, and it cancels in the sum: only the box lets energy into the pair.

Check. Every term is in watts. With a perfect box, $\dot{Q}_{\mathrm{box}} = 0$, (6) says $U_B + U_D$ stays constant: what the drink loses, the bath gains.

Where it fails. If mass crosses the boundary — the lid is lifted to add ice, or water evaporates — the systems are open, and the energy that mass carries must be added. A stirrer would add its work; there is none here.

3 Step 2: heat entering through Styrofoam

The idea. Heat flows from the warm room to the cold bath through the foam. Foam conducts poorly, so the flow is small, and it is proportional to the temperature difference.

Assumptions. The foam wall is thin compared with its width, so heat flows straight through it (one dimension). The foam stores no heat, so the flow is steady (B5). Its conductivity k is uniform. And, as in the first edition, its faces are at T_{room} and T.

Derivation. Fourier's law says that heat flows down a temperature gradient in proportion to it. Take a coordinate x through the wall from the bath side ($x = 0$) to the room side ($x = d$); the heat flux toward the bath is $q = k\,\mathrm{d}T/\mathrm{d}x$ ($\mathrm{W\,m^{-2}}$). If the foam stores no heat, whatever enters a slice of it leaves the other side, so q is the same at every x. Then $\mathrm{d}T/\mathrm{d}x$ is constant, and the temperature rises in a straight line from T to T_{room}:

$$\frac{\mathrm{d}T}{\mathrm{d}x} = \frac{T_{\mathrm{room}} - T}{d}, \qquad q = \frac{k}{d}\,(T_{\mathrm{room}} - T).$$

Over an area A the heat flow is qA. Adding it up over time, naming the conductance, and using

(3) for the bath alone:

$$\dot{Q}_{\text{Styrofoam}}(t) = \frac{kA}{d}\,(T_{\text{room}} - T(t)), \tag{7}$$

$$Q_{\text{in}}(t) = \int_0^t \frac{kA}{d}\,[T_{\text{room}} - T(\tau)]\,\mathrm{d}\tau, \tag{8}$$

$$H = \frac{kA}{d}, \tag{9}$$

$$\frac{\mathrm{d}U}{\mathrm{d}t} = H\,(T_{\text{room}} - T(t)). \tag{10}$$

H, in $\mathrm{W\,K^{-1}}$, is the heat flow for each kelvin of difference. Walls, base and lid of different thickness d_j and area A_j are separate paths side by side, so their flows add:

$$\dot{Q}_{\text{in}} = \sum_j \frac{k_j A_j}{d_j}\,(T_{\text{room}} - T). \tag{11}$$

Air films (correction). The foam's faces are not at T_{room} and T: still air on each side adds a film resistance $1/(hA)$ in series with the foam's $d/(kA)$. Resistances in series add, because the temperature drops add while the heat flow is the same through each:

$$\frac{1}{H_j} = \frac{1}{h_{\text{out}} A_j} + \frac{d_j}{k_j A_j} + \frac{1}{h_{\text{in}} A_j}. \tag{12}$$

Still room air gives $h_{\text{out}} \approx 9\,\mathrm{W\,m^{-2}K^{-1}}$ (about 5 by convection and 4 by radiation), and the air gap inside the box about 6. For 2 cm of foam with $k = 0.033$, $d/k = 0.61\,\mathrm{m^2K\,W^{-1}}$ while the films add $1/9 + 1/6 = 0.28$: the films cut the leak by about 30 %, and by 25 % for the rig's 3 cm.

Is the foam steady? A temperature change crosses a layer of thickness d in a time of order d^2/κ_f, where $\kappa_f = k/(\rho_f c_f)$ is the foam's thermal diffusivity. For expanded polystyrene, $\kappa_f = 0.035/(20 \times 1300) = 1.35 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$: 300 s for 2 cm, 670 s for 3 cm. After the first ten minutes or so the foam is steady. Before that it is still giving up heat it held at room temperature, which (7) leaves out.

Worked example: the leak

Demonstration: $H = 0.033 \times 0.05/0.02 = 0.0825\,\mathrm{W\,K^{-1}}$; with the bath at −10 °C and the room at 25 °C, $\dot{Q} = 0.0825 \times 35 = 2.9\,\mathrm{W}$. Rig: 30 mm of foam gives about $0.46\,\mathrm{W\,K^{-1}}$ alone and about 0.34 with its films and the gaps round the lid; with the bath at −21 °C and the room at 28 °C, 17 to 23 W.

Where it fails. Corners conduct more than flat walls, gaps round the lid let heat bypass the foam, and the foam's own heat matters early on. Measuring H directly, with the box test of the B8 README, is more reliable than computing it.

4 Step 3: the ice and the latent heat

The idea. Heat reaching the bath either warms what is there (sensible heat) or melts ice at constant temperature (latent heat).

Derivation. The specific heat c is the heat that warms 1 kg by 1 K, so warming a mass m_i of ice by $\mathrm{d}T$ takes $m_i c_i\,\mathrm{d}T$; dividing by $\mathrm{d}t$ gives the rate. The latent heat L_f is the heat that melts 1 kg of ice without changing its temperature, so melting $\mathrm{d}m_{\text{melt}}$ takes $L_f\,\mathrm{d}m_{\text{melt}}$. Ice that melts is ice

lost, $\dot{m}_{\text{melt}} = -\dot{m}_i$:

$$\mathrm{d}Q_i = m_i c_i \,\mathrm{d}T, \tag{13}$$
$$\dot{Q}_{i,\text{sensible}} = m_i c_i \,\dot{T}, \tag{14}$$
$$\mathrm{d}Q_{\text{melt}} = L_f \,\mathrm{d}m_{\text{melt}}, \tag{15}$$
$$\dot{Q}_{\text{melt}} = L_f \,\dot{m}_{\text{melt}}, \tag{16}$$
$$\dot{m}_{\text{melt}} = -\dot{m}_i, \tag{17}$$
$$\dot{Q}_{\text{melt}} = -L_f \,\dot{m}_i. \tag{18}$$

The heat entering through the box warms the ice, warms the liquid water ($m_w c_w \dot{T}$, by the same reasoning) and melts ice. Adding the three gives the first edition's combined balance:

$$H(T_{\text{room}} - T) = (m_i c_i + m_w c_w)\,\dot{T} - L_f \,\dot{m}_i. \tag{19}$$

When ice melts, $\dot{m}_i < 0$, so $-L_f \dot{m}_i$ is positive: melting absorbs heat.

Check. Pure ice and water coexist only at 0 °C, so there $\dot{T} = 0$ and (19) gives $\dot{m}_i = -H(T_{\text{room}} - T)/L_f$. Demonstration: $0.0825 \times 25/333\,500 = 6.2 \times 10^{-6}\,\mathrm{kg\,s^{-1}}$, 22 g of ice an hour.

Where it fails. L_f is the latent heat at 0 °C. Ice melting at −15 °C takes less, by $c_w - c_i$ for each kelvin below zero (Section 14). Ice straight from a freezer at −18 °C must first be warmed: $2100 \times 18 = 37.8\,\mathrm{kJ\,kg^{-1}}$, 11 % of L_f.

5 Step 4: liquid water and mass conservation

The idea. Water only changes form, ice to liquid or back, so ice and liquid always add up to the same total.

Derivation. The liquid's sensible heat follows exactly as the ice's:

$$Q_w = m_w c_w \,\Delta T, \tag{20}$$
$$\dot{Q}_{w,\text{sensible}} = m_w c_w \,\dot{T}. \tag{21}$$

If melting and freezing are the only changes of form, and no water enters or leaves (B6), every kilogram of ice lost is a kilogram of liquid gained. Integrating that from the start:

$$\mathrm{d}m_w = -\mathrm{d}m_i, \tag{22}$$
$$\dot{m}_w = -\dot{m}_i, \tag{23}$$
$$m_w(t) + m_i(t) = m_{w,0} + m_{i,0} = M_w, \tag{24}$$
$$m_w(t) = M_w - m_i(t). \tag{25}$$

Substituting (25) into (19) leaves the ice as the only mass:

$$H(T_{\text{room}} - T) = \left[m_i c_i + (M_w - m_i)\,c_w\right]\dot{T} - L_f \,\dot{m}_i. \tag{26}$$

Where it fails. Dissolved salt makes the liquid heavier than its water but is given no heat capacity here (B4): the brine's heat capacity is taken as that of its water.

6 Step 5: enthalpy of dissolution of NaCl

The idea. Dissolving salt takes heat: pulling the crystal apart costs more energy than the water molecules give back by gathering round the ions. A bath cools itself a little by dissolving its salt.

Derivation. A mass m_s of dissolved salt is $n_s = m_s/M_{\mathrm{NaCl}}$ moles. The heat of solution ΔH_{sol} is measured per mole, so dissolving at a rate $\dot{n}_s$ absorbs $\Delta H_{\mathrm{sol}}\,\dot{n}_s$:

$$n_s = \frac{m_s}{M_{\mathrm{NaCl}}}, \tag{27}$$

$$\dot{Q}_{\mathrm{sol}} = \Delta H_{\mathrm{sol}}\,\dot{n}_s = \frac{\Delta H_{\mathrm{sol}}}{M_{\mathrm{NaCl}}}\,\dot{m}_s = \beta\,\dot{m}_s, \tag{28}$$

$$\beta = \frac{3900}{0.05844} = 66\,735\ \mathrm{J\,kg^{-1}}.$$

Dissolving is endothermic, $\beta > 0$: a positive $\dot{m}_s$ takes heat, so it joins the other heat sinks on the right of (19):

$$\boxed{H(T_{\mathrm{room}} - T) = (m_i c_i + m_w c_w)\,\dot{T} - L_f\,\dot{m}_i + \frac{\Delta H_{\mathrm{sol}}}{M_{\mathrm{NaCl}}}\,\dot{m}_s} \tag{29}$$

or, eliminating m_w with (25),

$$H(T_{\mathrm{room}} - T) = \left[m_i c_i + (M_w - m_i)\,c_w\right]\dot{T} - L_f\,\dot{m}_i + \beta\,\dot{m}_s. \tag{30}$$

Worked example: dissolving

Dissolving 1 kg of salt absorbs 66.7 kJ, as much as melting 0.20 kg of ice. The demonstration's 50 g absorbs 3.3 kJ, the heat of melting 10 g of ice.

The salt's own heat capacity (correction). Undissolved salt also warms and cools with the bath, at $c_s = 864\,\mathrm{J\,kg^{-1}K^{-1}}$, and (29) leaves it out. To include it, add $(S_0 - m_s)\,c_s$ to the heat capacity in front of $\dot{T}$. For the rig, 1 kg of salt adds $864\,\mathrm{J\,K^{-1}}$ to the $15\,\mathrm{kJ\,K^{-1}}$ of its ice and water: 6 %. Steps 6–11 follow the first edition and leave it out; the equilibrium of Step 12 includes it, with the heat the salt brings in from the room.

Where it fails. ΔH_{sol} is measured in dilute solution at 25 °C, and it changes with concentration and temperature: one number is an approximation.

7 Step 6: freezing-point depression $i_v K_f b$

The idea. Ice melts in brine below 0 °C because the dissolved ions dilute the water: ice and brine can only coexist at a lower temperature than ice and pure water. While both are present the bath sits at that temperature (B2), so the salt in the brine fixes the bath temperature.

Where the law comes from. Ice and liquid water exchange molecules until water has the same *chemical potential* — Gibbs energy per mole — in both. Ice is pure water. In the brine the ions lower water's chemical potential below that of pure water, μ_w^*, by $R\mathcal{T}\ln a_w$, where $a_w < 1$ is the water's activity, $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$ and $\mathcal{T}$ the absolute temperature:

$$\mu_{\mathrm{ice}}(\mathcal{T}) = \mu_w^*(\mathcal{T}) + R\mathcal{T}\ln a_w.$$

At $\mathcal{T}_0 = 273.15\,\mathrm{K}$ pure ice and water coexist. The Gibbs–Helmholtz relation, with the molar heat of melting $\Delta h_{\mathrm{fus}} = L_f M_{\mathrm{H_2O}} = 333\,500 \times 0.018015 = 6008\,\mathrm{J\,mol^{-1}}$, turns the condition into

$$\ln a_w = -\frac{\Delta h_{\mathrm{fus}}}{R}\left(\frac{1}{\mathcal{T}} - \frac{1}{\mathcal{T}_0}\right).$$

In dilute brine, $\ln a_w$ is minus the moles of dissolved particles per mole of water. Each formula unit of NaCl gives $i_v = 2$ ions, and b moles of NaCl are dissolved in each kilogram of water,

which is $1/M_{\mathrm{H_2O}}$ moles of water, so $\ln a_w \approx -i_v\, b\, M_{\mathrm{H_2O}}$. For a small drop $\Delta T_f = \mathcal{T}_0 - \mathcal{T}$, $1/\mathcal{T} - 1/\mathcal{T}_0 = \Delta T_f/(\mathcal{T}\mathcal{T}_0) \approx \Delta T_f/\mathcal{T}_0^2$. Equating the two expressions for $\ln a_w$:

$$\Delta T_f = \underbrace{\frac{R\mathcal{T}_0^2 M_{\mathrm{H_2O}}}{\Delta h_{\mathrm{fus}}}}_{K_f}\, i_v\, b, \tag{31}$$

$$K_f = \frac{8.314 \times 273.15^2 \times 0.018015}{6008} = 1.86\ \mathrm{K\,kg\,mol^{-1}}.$$

The ions are not quite independent: the osmotic coefficient, about 0.93 between 0.1 and 1 $\mathrm{mol\,kg^{-1}}$, corrects for that, and in strong brine the logarithm can no longer be expanded.

Derivation. With b in moles of NaCl per kilogram of *liquid water*:

$$\Delta T_f = i_v K_f b, \tag{32}$$

$$b(t) = \frac{m_s(t)/M_{\mathrm{NaCl}}}{m_w(t)}, \tag{33}$$

$$T_f(t) = -i_v K_f b(t) = -\frac{i_v K_f\, m_s(t)}{M_{\mathrm{NaCl}}\, m_w(t)}, \tag{34}$$

$$T(t) = T_f(t) \quad \text{only when ice and brine coexist in equilibrium.} \tag{35}$$

(33) is moles of salt over kilograms of water. (34) is the freezing point in °C, minus the drop. (35): the bath sits at the freezing point only while there is ice to melt or freeze. Substituting the mass balance (25) and the shortcut $\alpha = i_v K_f/M_{\mathrm{NaCl}} = 63.65\,\mathrm{K}$:

$$\boxed{T = -\frac{i_v K_f\, m_s}{M_{\mathrm{NaCl}}\,(M_w - m_i)} = -\alpha\,\frac{m_s}{M_w - m_i}} \tag{36}$$

The bath gets colder as salt dissolves, and warmer as ice melts and dilutes the brine.

Worked example: the first edition's example, and how accurate it is

0.005 kg of salt dissolved in 0.100 kg of water:

$$T_f = -2 \times 1.86 \times \frac{0.005/0.05844}{0.100} = -3.18\,^{\circ}\mathrm{C}. \tag{37}$$

The measured freezing curve (Bodnar 1993) gives −2.87 °C for this brine, 4.76 wt % salt: the dilute law is 11 % too cold. In strong brine the error changes sign:

Salt in the brine (wt %)	5	10	15	16.5	20	23.2
Dilute law, ΔT_f (K)	3.35	7.07	11.23	12.58	15.91	19.21
Measured (Bodnar), ΔT_f (K)	3.03	6.61	11.03	12.57	16.71	21.20

How much salt the brine can hold. Brine holds at most C_{sat} of salt; once it is saturated, extra salt stays solid. With ice and saturated brine together, (36) with $m_s/W = C_{\mathrm{sat}}/\rho_w$ gives the coldest the model's bath can get:

$$T_{E,\mathrm{model}} = -\alpha\,\frac{C_{\mathrm{sat}}}{\rho_w}. \tag{38}$$

With halite's solubility, 357 kg per $\mathrm{m^3}$ of water (35.7 g per 100 g), this is −22.7 °C. The real coldest point, the eutectic, is −21.2 °C at 23.2 wt % (0.302 kg of salt per kg of water), where the dilute law gives −19.2 °C.

Correction. The dilute law (32) is good to about 10 % up to 15 wt %, but at the eutectic composition it is 2 K too warm, and with halite's solubility it puts the bath's floor at −22.7 °C instead of −21.2 °C. For strong brine, replace it with the measured freezing curve, $W = 1.78\,\vartheta - 0.0442\,\vartheta^2 + 5.572\times10^{-4}\,\vartheta^3$ (wt % of salt against ϑ degrees below zero; Bodnar 1993), and the measured eutectic.

8 Step 7: salt dissolution rate and shrinking crystals

The idea. Salt dissolves from the surface of each grain. Right at the surface the brine is saturated; further out it is weaker, and dissolved salt diffuses across the thin layer between. The rate is the area of the grains times how fast salt crosses that layer.

Derivation. Fick's law: salt diffuses down its concentration gradient at a flux D times the gradient, with D the diffusion coefficient. Across a layer of thickness δ, from saturated brine at the surface (C_{sat}) to the bulk (C), the flux is $D(C_{\mathrm{sat}} - C)/\delta$ in $\mathrm{kg\,m^{-2}s^{-1}}$. Calling $k_m = D/\delta$ the mass-transfer coefficient and multiplying by the exposed area A_s:

$$\dot m_s = k_m A_s\,(C_{\mathrm{sat}} - C), \tag{39}$$

$$C(t) \approx \frac{\rho_w\, m_s(t)}{m_w(t)}, \tag{40}$$

$$\dot m_s \approx k_m A_s\Big[C_{\mathrm{sat}}(T) - \frac{\rho_w m_s}{m_w}\Big]. \tag{41}$$

(40) measures the concentration as salt per volume of water, $m_s/(m_w/\rho_w)$, close to salt per volume of brine while the brine is dilute. For NaCl, $D \approx 1.5\times10^{-9}\,\mathrm{m^2s^{-1}}$; a layer 0.1 mm thick, as when brine flows past the grains, gives $k_m \approx 10^{-5}\,\mathrm{m\,s^{-1}}$, and a still layer ten times thicker 10^{-6}.

Shrinking crystals. Take N grains of the same shape, each a cube of side a. Their mass is $m_{\mathrm{solid}} = N\rho_{\mathrm{salt}}a^3$ and their area $6Na^2$. Eliminating a, the area is $6N^{1/3}(m_{\mathrm{solid}}/\rho_{\mathrm{salt}})^{2/3}$: for a fixed number of similar grains it goes as the solid mass to the power $\frac{2}{3}$, whatever their shape. With no salt dissolved at the start:

$$m_{\mathrm{solid}}(t) = m_{s,0} - m_s(t), \tag{42}$$

$$A_s(t) = A_{s,0}\Big(\frac{m_{s,0} - m_s(t)}{m_{s,0}}\Big)^{2/3}, \tag{43}$$

so

$$\boxed{\dot m_s = k_m A_{s,0}\Big(\frac{m_{s,0} - m_s}{m_{s,0}}\Big)^{2/3}\Big[C_{\mathrm{sat}}(T) - \frac{\rho_w m_s}{m_w}\Big]} \tag{44}$$

Worked example: how fast the rig's salt dissolves

1 kg of grains 0.5 mm across is 3.7×10^6 cubes, with $A_{s,0} = 5.5\,\mathrm{m^2}$. In fresh water, with $C_{\mathrm{sat}} = 357\,\mathrm{kg\,m^{-3}}$, the starting rate $k_m A_{s,0} C_{\mathrm{sat}}$ is $20\,\mathrm{g\,s^{-1}}$ for $k_m = 10^{-5}\,\mathrm{m\,s^{-1}}$ (the lot in about a minute, were the rate to stay up) and $2\,\mathrm{g\,s^{-1}}$ for 10^{-6} (about eight minutes). Both slow as the grains shrink and the brine strengthens.

Where it fails (from the first edition). It needs $0 \le m_s < m_{s,0}$: undissolved salt must remain, and the brine must be undersaturated. An extension that forbids precipitation uses $\max(0, C_{\mathrm{sat}} - C)$ and stops at complete dissolution. With salt dissolved at the start, use $S_0 - m_s$ in place of $m_{s,0} - m_s$. For concentrated brine use $C = m_s/V_{\mathrm{brine}}$ with the measured brine density. k_m and $A_{s,0}$ depend on grain size, on how brine moves through the ice and on whether a crust forms, and are best fitted to a run.

9 Step 8: the can and its drink (new)

The idea. The drink and its can are a second body in the bath. It has one temperature T_d and a heat capacity C_d, and it gives heat to the bath through the can wall at a rate proportional to $T_d - T$.

9.1 The drink as one body

Derivation. Apply (5) to the drink and can. With one temperature (C1) and constant specific heats, their energy measured from a reference temperature is $U_D = (m_d c_d + m_c c_c)(T_d - T_{\mathrm{ref}})$; the reference drops out on differentiating:

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -\dot{Q}_d, \qquad C_d = m_d c_d + m_c c_c. \tag{45}$$

The gas in the headspace is left out (C3). A 41 mL headspace of CO_2 at two atmospheres is $pV/(R\mathcal{T}) \approx 0.0035\,\mathrm{mol}$, 0.15 g, holding $0.13\,\mathrm{J\,K^{-1}}$: a ten-thousandth of C_d.

Worked example: Can 2

For a sugary soft drink, $C_d = 0.330 \times 4000 + 0.013 \times 900 = 1320 + 12 = 1332\,\mathrm{J\,K^{-1}}$. The can adds 1 %.

9.2 The heat path through the wall

Derivation. Heat leaves the drink through three layers in series: the film of drink on the inside of the wall, the aluminium, and the film of brine on the outside. By C2 none of them stores heat, so the same heat flow crosses each, driven by the temperature drop across it. Per unit area a film passes $h\,\Delta T$ — Newton's law of cooling, which defines the film coefficient h — and the wall $k_w \Delta T / t_w$, by Fourier's law as in Step 2:

$$\frac{\dot{Q}_d}{A_c} = h_i\,(T_d - T_{wi}) = \frac{k_w}{t_w}\,(T_{wi} - T_{wo}) = h_o\,(T_{wo} - T),$$

with T_{wi} and T_{wo} the wall's inner and outer faces and A_c the area that passes heat (Step 9). Rearranging each to a temperature drop and adding them, the wall temperatures cancel:

$$\dot{Q}_d = G\,(T_d - T), \qquad G = h_{\mathrm{ov}} A_c, \qquad \frac{1}{h_{\mathrm{ov}}} = \frac{1}{h_i} + \frac{t_w}{k_w} + \frac{1}{h_o}. \tag{46}$$

The aluminium hardly matters: $t_w/k_w = 10^{-4}/160 = 6 \times 10^{-7}\,\mathrm{m^2K\,W^{-1}}$, against $1/h_i = 2.5 \times 10^{-3}$ and $1/h_o = 3.7 \times 10^{-3}$. A plastic sleeve label would add its own resistance in series.

9.3 The film coefficients

Derivation. Nothing stirs the drink or the bath, so the films are set by free convection. Next to the wall the liquid is ΔT warmer or colder than the rest, so lighter or denser by $\rho \beta_T \Delta T$, with β_T its thermal expansion coefficient, and buoyancy drives it along the wall. Balancing buoyancy against viscosity, and the heat carried along the wall against the heat conducted across the layer, gives a layer thickness $\delta_T \sim \mathcal{L}\,\mathrm{Ra}^{-1/4}$, where $\mathcal{L}$ is the height of the wall and $\mathrm{Ra} = g\beta_T \Delta T \mathcal{L}^3/(\nu\kappa)$ the Rayleigh number (ν and κ are the liquid's viscosity and thermal diffusivity). The film coefficient is about k/δ_T, so

$$h \propto \left(\frac{\Delta T}{\mathcal{L}}\right)^{1/4}. \tag{47}$$

The constant comes from experiments. This document takes $h_i = 400$ and $h_o = 270\,\mathrm{W\,m^{-2}K^{-1}}$ for an upright can at a 25 K difference (a study of bottles, and a fit to a measured cooling time), and multiplies both by 1.2 for a can lying on its side, whose relevant length is its 6.6 cm diameter instead of its 11.7 cm height: $(11.66/6.6)^{1/4} = 1.15$ from (47) alone. Then $h_{\mathrm{ov}} = 161$ upright and 193 lying. In a part-filled upright can the drink's wall is only as tall as the drink, so h_i is a few per cent larger.

9.4 Is one drink temperature enough?

A still drink would not be uniform: its Biot number, $h_i(V/A)/k = 400 \times 0.0129/0.58 \approx 9$, is far above the 0.1 below which conduction alone keeps a body uniform. It is uniform enough because free convection stirs it: the cold layer on the wall is about 1 mm thick and sinks, and the core is mixed. Cold drink does collect at the bottom, so the core stratifies slowly; a probe at the top of the drink and one at the bottom test C1.

9.5 The fixed-bath solution

Derivation. If the bath stays at T, (45) with (46) is $C_d\dot{T}_d = -G(T_d - T)$: the difference $T_d - T$ falls at a rate proportional to itself. Separating the variables and integrating from $T_{d,0}$, then solving for the time to reach a target T^*:

$$T_d(t) = T + (T_{d,0} - T)\,e^{-t/\tau}, \qquad \tau = \frac{C_d}{G}, \qquad t_{T^*} = \tau \ln \frac{T_{d,0} - T}{T^* - T}. \tag{48}$$

Worked example: if the whole can passed heat

The whole surface of Can 2 is $A_c = 2\pi r\ell + 2\pi r^2 = 242 + 68 = 310\,\mathrm{cm^2}$. Upright, $G = 161 \times 0.0310 = 5.0\,\mathrm{W\,K^{-1}}$ and $\tau = 1332/5.0 = 266\,\mathrm{s}$; from 28 °C to 5 °C in a bath at −18 °C, $t = 266 \ln(46/23) = 185\,\mathrm{s}$. Step 9 shows why less of the wall passes heat.

What it does to the bath. The drink's heat enters the bath: $G(T_d - T)$ joins $H(T_{\mathrm{room}} - T)$ on the left of the bath balance (29) (Step 11).

10 Step 9: a can that is not completely full (new)

The idea. A sealed can holds gas above its drink. Heat crosses the wall well only where there is liquid on both sides: drink inside, brine outside. The gas also decides whether the can floats, and how much room the drink has to expand if it freezes.

10.1 What the headspace changes

- **The drink's mass** is whatever is in the can: weigh it full and empty. The headspace does not change C_d (Step 8).
- **The wall with gas behind it passes almost no heat.** Heat would have to reach it through the gas, which conducts twenty to thirty times worse than water and barely convects in so thin a layer: a film coefficient of a few $\mathrm{W\,m^{-2}K^{-1}}$, a hundred times less than h_i. So the area that passes heat, A_c, is the wall wetted by drink inside *and* by brine outside (C4).
- **Conduction along the aluminium** carries a little heat past the drink's surface. The wall acts as a fin whose reach is $\lambda = \sqrt{k_w t_w / h_o} = \sqrt{160 \times 10^{-4}/270} = 7.7\,\mathrm{mm}$: a strip about that wide along the drink's surface passes some heat, adding up to about 6 % to G for the upright can. It is left out, so the model slightly underestimates G.

10.2 Measuring the fill

Stand the can upright and measure the headspace height h_g, from the top of the cylinder to the drink's surface; or fill an empty can with water to find its volume and subtract the drink's volume m_d/ρ_d. Treating the can as a cylinder of radius r and height ℓ, the gas volume is $V_g = \pi r^2 h_g$ and the fill fraction $f = 1 - h_g/\ell$. A real can's neck and domed base hold less than $\pi r^2 \ell$: Can 2's cylinder would hold 399 mL, against 330 mL of drink and about 41 mL of headspace. Measure volumes rather than compute them.

10.3 Upright can

Derivation. The drink stands $z_d = \ell - h_g$ high. Inside, the side wall is wet up to z_d and the base all over; the lid has gas behind it. Outside, the brine reaches a depth z_b up the can ($z_b \geq \ell$ when the can is held under). The wall is wet on both sides up to the lower of the two heights, so the side wall passes heat up to that height, and the base passes heat everywhere:

$$A_c = 2\pi r \, \min(z_d, z_b) + \pi r^2. \tag{49}$$

10.4 Lying can: the circular segment

Derivation. Lying on its side, the can's cross-section is a circle of radius r, and the drink fills the part below a horizontal chord: a circular *segment*. A chord at height y above the lowest point is $r - y$ below the centre, so it subtends a half-angle θ at the centre with $\cos\theta = (r-y)/r$. The segment is the sector of angle 2θ, of area $r^2\theta$, minus the triangle between the chord and the centre, of area $(r-y)\,r\sin\theta = (r-y)\sqrt{2ry - y^2}$; and the arc below the chord is $2r\theta$ long:

$$A_{\text{seg}}(y) = r^2 \arccos\frac{r-y}{r} - (r-y)\sqrt{2ry - y^2}, \tag{50}$$
$$P(y) = 2r \arccos\frac{r-y}{r}.$$

The formula holds for $y > r$ too, where $r - y$ is negative and the "triangle" adds to the sector. Laying the can down does not change the volume of its gas, so the gas strip along the top has the same volume as the upright headspace. That fixes the strip's depth y_g, and the drink's depth is what is left:

$$\ell\, A_{\text{seg}}(y_g) = \pi r^2 h_g, \qquad y_d = 2r - y_g. \tag{51}$$

With brine y_b up the can, the wall is wet on both sides below $y = \min(y_d, y_b)$: along the length of the can (the arc times ℓ) and on the two flat ends (a segment each):

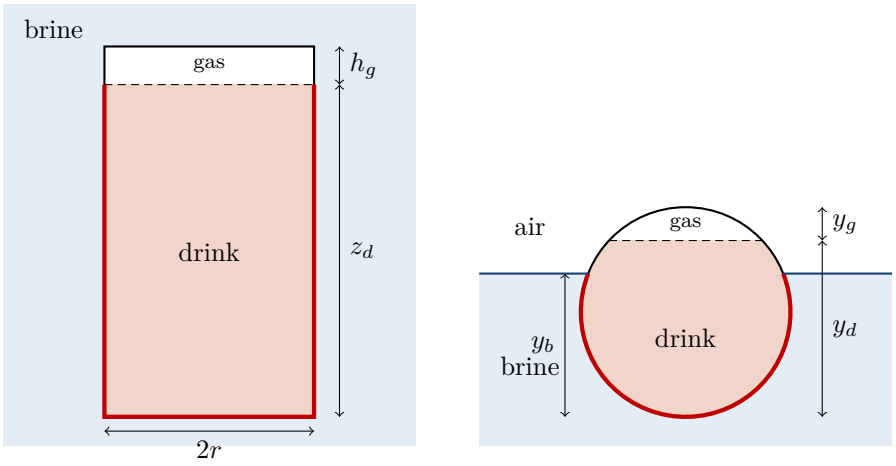
$$A_c = \ell\, P(y) + 2\, A_{\text{seg}}(y), \qquad y = \min(y_d, y_b). \tag{52}$$

Worked example: Can 2 held under the brine

$r = 3.3\,\text{cm}$, $\ell = 11.66\,\text{cm}$, $h_g = 1.2\,\text{cm}$. Upright: $z_d = 10.46\,\text{cm}$ ($f = 0.897$) and (49) gives $A_c = 217 + 34 = 251\,\text{cm}^2$, 81 % of the whole surface. Lying: (51) needs a segment of $\pi r^2 h_g/\ell = 3.52\,\text{cm}^2$, 10 % of the circle, so the gas strip is $y_g = 1.05\,\text{cm}$ deep and the drink $y_d = 5.55\,\text{cm}$; (52) gives $A_c = 178.5 + 61.4 = 240\,\text{cm}^2$, 77 %.

10.5 Does the can float?

Derivation. By Archimedes' principle, a floating body sinks until the liquid it displaces weighs as much as it does. The can and drink weigh $(m_d + m_c)g$; a volume V_{imm} of brine of density ρ_b



(a) upright, held under (b) lying, floating

Figure 1: Can 2 to scale, part-filled. (a) Upright and held under the brine: 1.2 cm of headspace above 10.46 cm of drink. (b) End-on, lying and floating in strong brine: a gas strip 1.05 cm deep, drink 5.55 cm deep, and the brine 4.51 cm up the can. Heat passes only where the wall has drink inside and brine outside (red).

weighs $\rho_b g V_{\text{imm}}$. So $V_{\text{imm}} = (m_d + m_c)/\rho_b$, and if that is less than the can's volume, the can floats. Upright, the immersed volume is $\pi r^2 z_b$; lying, it is $\ell\, A_{\text{seg}}(y_b)$:

$$z_b = \frac{m_d + m_c}{\rho_b\, \pi r^2}, \qquad A_{\text{seg}}(y_b) = \frac{m_d + m_c}{\rho_b\, \ell}. \tag{53}$$

For Can 2, 343 g in all, in strong brine ($\rho_b = 1182\,\text{kg m}^{-3}$): $z_b = 8.48$ cm upright, $y_b = 4.51$ cm lying. A part-filled can is lighter for its size, so it floats higher than a full one.

Which way up? Tilt a floating upright can by a small angle. The displaced brine shifts to the low side and moves the centre of buoyancy B sideways, as if the can hung from a point M, the *metacentre*, a height $BM = I/V_{\text{imm}}$ above B, where $I = \pi r^4/4$ is the second moment of the can's cross-section at the brine surface. The can rights itself if M lies above its centre of mass. Measuring from the bottom of the can, B is at $z_b/2$ and the centre of mass at about $z_d/2$, the middle of the drink, so the height of M above the centre of mass is

$$GM = \frac{z_b}{2} + \frac{r^2}{4 z_b} - \frac{z_d}{2}. \tag{54}$$

For Can 2, $GM = 4.24 + 0.32 - 5.25 = -0.69$ cm (the centre of mass at 5.25 cm includes the can). The drink sloshing inside, with a free surface of its own, lowers it further, by $(\rho_d/\rho_b)\, BM$, to -0.97 cm. $GM < 0$: an upright can floating in brine is unstable and **tips onto its side**. In fresh water it is marginal (+0.03 cm, and -0.25 cm with the drink's free surface) and falls over too. A can that is not held floats lying down, as in the rig.

Worked example: Can 2 floating on its side

The brine reaches $y_b = 4.51$ cm up a can 6.6 cm across, below the drink's 5.55 cm, so the wall is wet on both sides only below 4.51 cm: $A_c = 200\,\text{cm}^2$, 64 % of the surface. The top centimetre of drink faces the air in the box instead, which exchanges heat at only 5–10 $\text{W m}^{-2}\text{K}^{-1}$, and is left out.

10.6 What the fill and the immersion do to the cooling

Table 3 puts the areas into (46) and (48), for a bath held at $-18\,^\circ$C and a drink starting at $28\,^\circ$C.

Table 3: Can 2 with a cola, cooled from 28 °C to 5 °C in a bath held at −18 °C.

Can	A_c (cm^2)	G (W K^{-1})	τ (s)	time to 5 °C (s)
Whole surface passes heat (first estimate), upright	310	5.00	266	185
Part-filled, upright, held under	251	4.05	329	228
Part-filled, lying, held under	240	4.64	287	199
Part-filled, lying, floating	200	3.86	345	239

Ignoring the headspace makes an upright can cool about 20 % too fast. Lying down loses wetted area but gains stronger films, and ends only 8 % slower than the whole-surface estimate. Floating costs another 20 %: hold the can under.

10.7 Room to freeze

If the drink starts to freeze (Step 10), its ice takes more room than the water it came from: $1/917 - 1/1000 = 9.05 \times 10^{-5}\,\mathrm{m^3}$ per kg, about 0.09 mL per gram of ice. The headspace takes up the expansion by compressing its gas, whose pressure at constant temperature goes as $1/V$ (Boyle's law):

$$\frac{p}{p_0} = \frac{V_g}{V_g - \Delta V_{\mathrm{ice}}}, \qquad \Delta V_{\mathrm{ice}} = 9.05 \times 10^{-5}\, m_{\mathrm{ice}}. \tag{55}$$

240 g of ice, Step 10's cola at −5 °C, needs 22 mL. In a 41 mL headspace that more than doubles the pressure; in a real can's smaller headspace the can may bulge or burst.

11 Step 10: the drink as a solution, an ethanol model (new)

The idea. A drink is water with things dissolved in it: sugar, carbon dioxide and, in a beer or a cider, ethanol. They set two things: the drink's heat capacity c_d, which says how much heat must come out, and its freezing point T_{fd}, below which it starts to turn to slush. Dissolved molecules lower the freezing point exactly as salt lowers the bath's (Step 6), but each counts once, since none splits into ions.

11.1 Composition from the label

Derivation. Alcohol by volume, ABV, is the volume of ethanol per volume of drink. Per cubic metre of drink, the ethanol's mass is $\rho_e\,\mathrm{ABV}$, with $\rho_e = 789.2\,\mathrm{kg\,m^{-3}}$, and the drink's mass is ρ_d; their ratio is the ethanol's mass fraction. Sugar and CO_2 are given as mass per volume, s and c in $\mathrm{g\,L^{-1}}$ (equal to $\mathrm{kg\,m^{-3}}$); dividing by ρ_d gives their mass fractions. Beer also holds about 3.5 % of unfermented extract, w_x, mostly large molecules. The rest is water:

$$w_e = \frac{\rho_e\,\mathrm{ABV}}{\rho_d}, \qquad w_{su} = \frac{s}{\rho_d}, \qquad w_c = \frac{c}{\rho_d}, \tag{56}$$
$$w_{wd} = 1 - w_e - w_{su} - w_c - w_x.$$

The molality of each solute is its moles per kilogram of the drink's water: its mass fraction over its molar mass, divided by the water's mass fraction:

$$b_k = \frac{w_k}{M_k\, w_{wd}}, \qquad k = e,\ su,\ c. \tag{57}$$

11.2 Where the drink starts to freeze

Derivation. The derivation of (31) counted dissolved particles per kilogram of water, whatever they were. Ethanol, sugar and CO_2 stay whole molecules in water, $i_v = 1$, so their molalities simply add:

$$\boxed{T_{fd} = -K_f\,(b_e + b_{su} + b_c)} \tag{58}$$

The extract's molecules are large, so few per gram, and it is left out.

Table 4: Three drinks, from (56)–(58) and (62).

	Cola	Beer, 5 % ABV	Hard seltzer, 5 % ABV
Density ρ_d ($\mathrm{kg\,m^{-3}}$)	1040	1008	993
Sugar; CO_2 ($\mathrm{g\,L^{-1}}$)	106 (sucrose); 6.5	0; 5.0	0; 6.0
Ethanol w_e; water w_{wd}	0; 0.892	0.039; 0.921	0.040; 0.954
b_e, b_{su}, b_c ($\mathrm{mol\,kg^{-1}}$)	0, 0.334, 0.159	0.923, 0, 0.122	0.904, 0, 0.144
Freezing point T_{fd}	−0.92 °C	−1.94 °C	−1.95 °C
Specific heat c_d ($\mathrm{kJ\,kg^{-1}K^{-1}}$)	3.95	4.14	4.24

5 % of ethanol lowers the freezing point by 1.7 K, more than the 10 % of sugar in a cola (0.6 K): an ethanol molecule has a seventh of the mass of a sucrose molecule, so per gram it counts seven times. Measured freezing points of beers are close to −2 °C. Glucose–fructose syrup, whose molecules are half the mass of sucrose's, counts twice per gram. The law is the dilute one; for spirits, such as 40 % ABV, use measured freezing points.

11.3 Freezing gradually: slush

Derivation. Ice that forms in the drink is pure water; the ethanol, sugar and CO_2 stay in the liquid, which gets stronger as it loses water. If a share ψ of the drink's water has frozen, every molality is $1/(1-\psi)$ times its starting value, and by (58) so is the freezing-point drop. The liquid that is left is in equilibrium with the ice when

$$T_d = \frac{T_{fd}}{1-\psi} \qquad \Longleftrightarrow \qquad \psi(T_d) = 1 - \frac{T_{fd}}{T_d} \quad (T_d < T_{fd}). \tag{59}$$

Below T_{fd} the drink is a slush whose ice grows as it cools: half its water frozen at $2T_{fd}$, three quarters at $4T_{fd}$.

Freezing releases latent heat, which the bath must take away. The drink's energy, measured as in Step 8, gains a latent term for its ice; neglecting the small difference between the heat capacities of its ice and its water,

$$U_D = C_d\,(T_d - T_{\mathrm{ref}}) - L_f\,m_{wd}\,\psi(T_d), \tag{60}$$

with $m_{wd} = w_{wd} m_d$ the water in the drink. Differentiating with respect to T_d, with $\mathrm{d}\psi/\mathrm{d}T_d = T_{fd}/T_d^2$ from (59) (negative, since $T_{fd} < 0$: more ice as it cools), gives the heat that cools the drink by one kelvin, its effective heat capacity:

$$\boxed{\begin{aligned} &C_{d,\mathrm{eff}}(T_d) = C_d + \frac{L_f\,m_{wd}\,|T_{fd}|}{T_d^2} && (T_d < T_{fd}),\\ &C_{d,\mathrm{eff}}(T_d) = C_d && (T_d \geq T_{fd}) \end{aligned}} \tag{61}$$

and the drink's equation (45) becomes $C_{d,\mathrm{eff}}(T_d)\,\dot{T}_d = -\dot{Q}_d$.

Worked example: a beer and a cola left in

Beer: 304 g of water, $T_{fd} = -1.94\,^\circ\mathrm{C}$. Just below T_{fd}, $C_{d,\mathrm{eff}}$ jumps from $1.38\,\mathrm{kJ\,K^{-1}}$ to $53\,\mathrm{kJ\,K^{-1}}$, forty times larger, so the drink lingers near $-2\,^\circ\mathrm{C}$ while it freezes. At $-5\,^\circ\mathrm{C}$ it is 61 % frozen (186 g of ice, 17 mL of expansion); at $-17\,^\circ\mathrm{C}$, 89 %. A cola, $T_{fd} = -0.92\,^\circ\mathrm{C}$ with 294 g of water, is 82 % frozen at $-5\,^\circ\mathrm{C}$ (240 g of ice, 22 mL). Ethanol keeps a beer liquid a kelvin lower, but a drink left in a $-20\,^\circ\mathrm{C}$ bath ends mostly ice either way.

11.4 The drink's heat capacity

Derivation. A solution's heat capacity is a sum over what it contains, each part weighted by its mass fraction and counted with its *partial* specific heat: what a kilogram of it adds to the solution, which is not its heat capacity as a pure substance:

$$c_d = w_{wd}\,c_w + w_e\,\bar{c}_e + w_{su}\,\bar{c}_{su} + w_x\,\bar{c}_x. \tag{62}$$

In dilute solution ethanol adds about $5.4\,\mathrm{kJ\,kg^{-1}K^{-1}}$ ($0.25\,\mathrm{kJ\,mol^{-1}K^{-1}}$), more than twice pure ethanol's 2.44 and more than water's: water molecules order themselves round the ethanol's ethyl group, and that order melts as the solution warms, which takes heat. Sucrose adds about $1.9\,\mathrm{kJ\,kg^{-1}K^{-1}}$ and beer's extract about 1.5; the CO_2 is too little to matter. So a little ethanol *raises* c_d slightly above water's, while sugar lowers it: the values in Table 4. A mass-weighted average with pure ethanol's 2.44 would wrongly put a beer below water.

These are estimates, good to a few per cent. A drink's own value is easily measured by mixing: pour a mass m_1 of drink at T_1 into a mass m_2 of water at T_2 in an insulated cup and read the final temperature T_3. The heat the drink loses is the heat the water gains, so $c_d = m_2 c_w (T_3 - T_2)/[m_1(T_1 - T_3)]$.

11.5 Convection inside the can

At 5 % ethanol the drink is about 15 % more viscous than water and slightly less conductive; through the quarter powers of Step 8 these change h_i by a few per cent. More important near the end of a run: water is densest at $4\,^\circ\mathrm{C}$, and as the drink at the wall approaches that temperature it stops getting denser, so the free convection that mixes it weakens. Dissolved sugar and CO_2 move that point lower. The usual target, $5\,^\circ\mathrm{C}$, is just above it.

11.6 Supercooling

A still drink in a smooth can often cools several kelvin below T_{fd} before any ice appears, then freezes part of the way within seconds, its latent heat warming it back onto the curve (59). The model assumes ice appears at T_{fd}.

11.7 What ethanol changes

In the coupled model of Step 11 a beer takes 3 % longer than a cola to reach $5\,^\circ\mathrm{C}$ (the worked example of Step 11), because its c_d is larger. Ethanol matters little for the cooling time; it matters for what happens after, by keeping the drink liquid a kelvin lower.

11.8 Ethanol in the bath instead of salt

The same law describes ice with ethanol in place of salt, with $i_v = 1$ and $M_e = 0.04607\,\mathrm{kg\,mol^{-1}}$: per unit of the mass ratio m_s/W, ethanol lowers the freezing point by $K_f/M_e = 40.4\,\mathrm{K}$ against

salt's 63.7 K, so in dilute solution salt is the stronger by weight. Two things differ more. Ethanol mixes with water in any proportion, so there is no saturation and no eutectic plateau within reach: such a bath has no thermostat, and warms as its ice melts. And mixing ethanol with water releases heat, where dissolving salt absorbs it. This document does not model an ethanol bath further.

12 Step 11: fully unified time-dependent model, with the can

The idea. Put the pieces together: the bath's energy balance (Step 5) with the drink's heat in it, the freezing point (Step 6) tying the bath temperature to its salt and water, the dissolution rate (Step 7), and the drink's equation (Steps 8 and 10). Then solve for the rates.

Shortcuts (the first edition's, and the can's):

$$W = M_w - m_i = m_w, \tag{63}$$

$$C_{\text{eff}} = m_i c_i + W c_w, \tag{64}$$

$$\alpha = \frac{i_v K_f}{M_{\text{NaCl}}}, \qquad \beta = \frac{\Delta H_{\text{sol}}}{M_{\text{NaCl}}}, \qquad H = \frac{kA}{d}, \qquad G = h_{\text{ov}} A_c. \tag{65}$$

α is in kelvin per unit of the mass ratio m_s/W, and β in $\mathrm{J\,kg^{-1}}$.

Derivation. The bath now receives heat from the room and from the drink. Call their sum

$$\dot{Q}_{\text{in}} = H(T_{\text{room}} - T) + G(T_d - T); \tag{66}$$

it replaces the left side of (29), whose right side is unchanged. The freezing point is (36):

$$\dot{Q}_{\text{in}} = C_{\text{eff}}\,\dot{T} - L_f\,\dot{m}_i + \beta\,\dot{m}_s, \tag{67}$$

$$T = -\alpha\,\frac{m_s}{W}. \tag{68}$$

Differentiate (68) with the quotient rule, using $\dot{W} = -\dot{m}_i$ from (63):

$$\dot{T} = -\alpha\,\frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{m_s}{W}\Big) \tag{69}$$

$$= -\alpha\,\frac{\dot{m}_s W - m_s \dot{W}}{W^2} \tag{70}$$

$$= -\alpha\Big(\frac{\dot{m}_s}{W} + \frac{m_s\,\dot{m}_i}{W^2}\Big). \tag{71}$$

Dissolving salt cools the bath (the first term); melting ice warms it, since $\dot{m}_i < 0$ (the second). Substituting (71) into (67), then collecting the terms in $\dot{m}_s$ and in $\dot{m}_i$:

$$\dot{Q}_{\text{in}} = -\alpha C_{\text{eff}}\Big(\frac{\dot{m}_s}{W} + \frac{m_s\,\dot{m}_i}{W^2}\Big) - L_f\,\dot{m}_i + \beta\,\dot{m}_s \tag{72}$$

$$= \Big(\beta - \frac{\alpha C_{\text{eff}}}{W}\Big)\dot{m}_s - \Big(L_f + \frac{\alpha C_{\text{eff}} m_s}{W^2}\Big)\dot{m}_i. \tag{73}$$

Solving for the ice, and writing $\dot{Q}_{\text{in}}$ out again:

$$\boxed{\dot{m}_i = \frac{\Big(\beta - \dfrac{\alpha C_{\text{eff}}}{W}\Big)\dot{m}_s - H(T_{\text{room}} - T) - G(T_d - T)}{L_f + \dfrac{\alpha C_{\text{eff}} m_s}{W^2}}} \tag{74}$$

What the terms mean. The denominator is the heat it takes to melt a kilogram of ice *in this bath*: L_f, plus the heat that warms the whole bath, because the meltwater dilutes the brine

and raises its freezing point. In the numerator, heat from the room and from the drink melts ice. Dissolving salt melts ice too, through the bracket, which is negative in practice: the salt lowers the freezing point, the bath must cool to follow it, and the heat for that cooling comes from melting ice. This is how a salted ice bath cools itself.

Worked example: the rig bath as the can goes in

After 15 minutes with no stirrer (the run below): $m_s = 0.633\,\mathrm{kg}$ and $T = -16.8\,^\circ\mathrm{C}$, so $W = \alpha m_s/|T| = 2.40\,\mathrm{kg}$, $m_i = 3.20\,\mathrm{kg}$ and $C_{\mathrm{eff}} = 16.8\,\mathrm{kJ\,K^{-1}}$. The denominator is $333\,500 + 116\,800 = 450\,300\,\mathrm{J\,kg^{-1}}$, 35 % more than L_f. The bracket is $66\,700 - 443\,800 = -377\,100\,\mathrm{J\,kg^{-1}}$: each kilogram of salt that dissolves melts $377\,100/450\,300 = 0.84\,\mathrm{kg}$ of ice.

The drink. From Steps 8 and 10:

$$C_{d,\mathrm{eff}}(T_d)\,\dot{T}_d = -G\,(T_d - T). \tag{75}$$

The full reduced system.

$$\boxed{\begin{aligned}
\dot{m}_s &= k_m A_{s,0}\Big(\frac{m_{s,0} - m_s}{m_{s,0}}\Big)^{2/3}\Big[C_{\mathrm{sat}}(T) - \frac{\rho_w m_s}{W}\Big],\\
\dot{m}_i &= \frac{(\beta - \frac{\alpha C_{\mathrm{eff}}}{W})\dot{m}_s - H(T_{\mathrm{room}} - T) - G(T_d - T)}{L_f + \frac{\alpha C_{\mathrm{eff}} m_s}{W^2}},\\
\dot{T}_d &= -\frac{G\,(T_d - T)}{C_{d,\mathrm{eff}}(T_d)},\\
T &= -\alpha\,\frac{m_s}{W}, \qquad W = M_w - m_i, \qquad C_{\mathrm{eff}} = m_i c_i + W c_w.
\end{aligned}} \tag{76}$$

Three state variables, m_s, m_i and T_d, move by the three rate equations; the bath temperature follows from the last line. Start with $m_i(0)$ and $m_s(0)$ consistent with (68), and $T_d(0)$ the drink's starting temperature; before the can goes in, set $G = 0$. This *ice–brine equilibrium branch* stops applying when the ice runs out, when the brine saturates (then $\dot{m}_s = 0$ and the bath sits near (38)), or when the bath is far from phase equilibrium.

Alternative single implicit equation. Eliminating T from the heat balance with (68), without solving for $\dot{m}_i$:

$$\begin{aligned}
&H\Big(T_{\mathrm{room}} + \frac{\alpha m_s}{W}\Big) + G\Big(T_d + \frac{\alpha m_s}{W}\Big)\\
&\qquad = -\alpha C_{\mathrm{eff}}\Big(\frac{\dot{m}_s}{W} + \frac{m_s\,\dot{m}_i}{W^2}\Big) - L_f\,\dot{m}_i + \beta\,\dot{m}_s.
\end{aligned} \tag{77}$$

It is equivalent to (74) paired with (68).

Worked example: the rig with Can 2, solved

(76) integrated numerically with the rig's numbers: a box of $0.46\,\mathrm{W\,K^{-1}}$ and a room at $28\,^\circ\mathrm{C}$; the $4\,^\circ\mathrm{C}$ water first melts 80 g of ice so that the bath starts at $0\,^\circ\mathrm{C}$; grains 0.5 mm across, $A_{s,0} = 5.5\,\mathrm{m^2}$; $C_{\mathrm{sat}} = 357\,\mathrm{kg\,m^{-3}}$ held constant; the can goes in at $28\,^\circ\mathrm{C}$ after 15 minutes.

Can	G (W K^{-1})	time to 5 °C (s)	
		$k_m = 10^{-6}$	$k_m = 10^{-5}$
Whole surface passes heat, upright	5.00	192	165
Part-filled, upright, held under	4.05	235	202
Part-filled, lying, held under	4.64	207	176
Part-filled, lying, floating	3.86	246	213
As above, with a beer	4.64	213	183
Bath when the can goes in		−16.8 °C	−22.4 °C
Salt dissolved by then		0.63 kg	0.93 kg

k_m is in m s^{-1}. With still brine ($k_m = 10^{-6}$) the salt dissolves slowly and the bath is only at −16.8 °C after 15 minutes; with brine moving past the grains (10^{-5}) it has nearly reached the model's floor, −22.7 °C. A drink left in for two hours ends at −19.2 °C: a cola 95 % frozen, a beer 90 %.

13 Step 12: time-independent equilibrium, with the can

The idea. There are two different things "time-independent" can mean: energy accounting between two moments, and a true steady state that nothing changes any more. They are not interchangeable.

13.1 Insulated equilibrium

Derivation. With no heat in, (2) says the energy at the end equals the energy at the start, $U_f = U_0$. Measure energies from ice and liquid water at $T_0 = 0$ °C, with the first edition's reference: liquid water holds L_f per kilogram more than ice. Between start and end, ice and water warm or cool to T_f (sensible heat), $m_{i,0} - m_{i,f}$ of ice melts (latent energy gained) and $m_{s,f} - m_{s,d,0}$ of salt dissolves (heat of solution absorbed); their sum is zero:

$$\boxed{0 = (m_{i,f} c_i + m_{w,f} c_w)\, T_f - L_f\,(m_{i,f} - m_{i,0}) + \beta\,(m_{s,f} - m_{s,d,0})} \tag{78}$$

With ice and brine together at the end, the mass balance and the freezing point close it:

$$m_{w,f} = M_w - m_{i,f}, \tag{79}$$

$$T_f = -\alpha\,\frac{m_{s,f}}{M_w - m_{i,f}}. \tag{80}$$

If solid salt remains, the brine is saturated, which is a third condition:

$$C_f = C_{\mathrm{sat}}(T_f); \tag{81}$$

if all the salt has dissolved, $m_{s,f} = S_0$ instead. These are alternative regimes, not simultaneous conditions: try one, and check that its answer is consistent.

With the can, and other starting temperatures (new). The drink cools from $T_{d,0}$ to T_f; if $T_f < T_{fd}$ part of its water freezes, by (59), giving up latent heat. Water starting at $T_{w,0}$ and salt at $T_{s,0}$ bring their own sensible heat, measured from 0 °C, on the starting side; the salt that is still solid at the end is counted at T_f (Step 5's correction). With the ice starting at 0 °C:

$$\begin{aligned} 0 = {} & \left[m_{i,f} c_i + m_{w,f} c_w + (S_0 - m_{s,f})\, c_s\right] T_f - L_f\,(m_{i,f} - m_{i,0}) \\ & + \beta\,(m_{s,f} - m_{s,d,0}) + C_d\,(T_f - T_{d,0}) - L_f\, m_{wd}\, \psi(T_f) \\ & - m_{w,0} c_w T_{w,0} - S_0 c_s T_{s,0}. \end{aligned} \tag{82}$$

Worked example: the rig and Can 2, left until everything settles

Rig numbers, with the drink at 28 °C; "measured diagram" means the freezing curve, solubility and eutectic of the B8 report's model (`b8_sim.py`) instead of (80)–(81).

	T_f	ice left	solid salt left	drink frozen
Bath alone, dilute law	−22.7 °C	2.85 kg	19 g (saturated)	—
Bath alone, measured diagram	−21.2 °C	2.93 kg	193 g	—
With a cola, dilute law	−20.3 °C	2.46 kg	0	95 %
With a cola, measured diagram	−21.2 °C	2.42 kg	40 g	96 %
With a beer, dilute law	−20.3 °C	2.46 kg	0	90 %
With a beer, measured diagram	−21.2 °C	2.42 kg	40 g	91 %

The drink gives the bath about 160 kJ, cooling from 28 °C and freezing, which melts 0.4–0.5 kg of ice. With the dilute law that dissolves the last of the salt and leaves the bath at −20.3 °C; with the measured diagram the bath stays on its eutectic plateau at −21.2 °C and uses 150 g of its spare salt. (A beer and a cola give almost the same heat: the beer's larger heat capacity is offset by its smaller frozen share.) Either way a can left until everything settles is mostly ice: take it out at 5 °C.

13.2 Integrated heat leakage over a finite interval

Derivation. Integrating the bath balance (67) from 0 to t_f, for constant c_i, c_w, L_f and β, the latent and dissolution terms integrate exactly, but the sensible term does not, because C_{eff} changes as ice melts:

$$\begin{aligned}\int_0^{t_f} \{H[T_{\text{room}} - T(t)] + G[T_d(t) - T(t)]\}\,\mathrm{d}t \\ = \int_0^{t_f} C_{\text{eff}}(t)\,\dot{T}(t)\,\mathrm{d}t - L_f(m_{i,f} - m_{i,0}) + \beta(m_{s,f} - m_{s,d,0}).\end{aligned} \tag{83}$$

The first edition's endpoint formula,

$$\begin{aligned}\int_0^{t_f} H[T_{\text{room}} - T(t)]\,\mathrm{d}t \\ \overset{\text{earlier approximation}}{\approx} C_f T_f - C_0 T_0 - L_f(m_{i,f} - m_{i,0}) + \beta(m_{s,f} - m_{s,d,0}),\end{aligned} \tag{84}$$

is not generally valid when C_{eff} varies.

Correction (first edition). Integrating by parts,

$$\int_0^{t_f} C_{\text{eff}}\,\dot{T}\,\mathrm{d}t = \left[C_{\text{eff}}T\right]_0^{t_f} - \int_0^{t_f} T\,\dot{C}_{\text{eff}}\,\mathrm{d}t, \qquad \dot{C}_{\text{eff}} = (c_i - c_w)\,\dot{m}_i, \tag{85}$$

so (83), not the endpoint replacement (84), follows the rate equation literally.

The drink's part, by contrast, integrates exactly: from (75), $G(T_d - T) = -C_{d,\text{eff}}\dot{T}_d$, so the heat it gives the bath between two moments is $C_d(T_{d,0} - T_d)$, plus $L_f m_{wd}\psi$ if it has started to freeze. Its heat capacity, unlike the bath's, does not change.

13.3 True steady state with heat leakage

Derivation. In a steady state nothing changes: $\dot{T} = \dot{m}_i = \dot{m}_s = \dot{T}_d = 0$. Then (75) gives $G(T_d - T) = 0$, and (67) gives $H(T_{\text{room}} - T) = 0$:

$$H(T_{\text{room}} - T_{\text{eq}}) = 0, \qquad G(T_{d,\text{eq}} - T_{\text{eq}}) = 0, \tag{86}$$

$$T_{\text{eq}} = T_{d,\text{eq}} = T_{\text{room}} \qquad (H > 0,\ G > 0). \tag{87}$$

Above 0 °C no ice can be in equilibrium, so the true steady state comes only after all the ice has melted; it cannot be found by imposing the ice–brine freezing condition on a warm room.

14 A thermodynamically more consistent refinement

The idea. The model so far treats L_f as a constant while giving ice and water different specific heats, $c_i \neq c_w$. That is inconsistent away from 0 °C. A consistent model starts from an energy that is a function of the state, and differentiates it.

Derivation. Choose $T_{\text{ref}} = 0\,°\text{C}$; give ice zero energy there, and liquid water L_f. In a first ideal-mixture approximation, the bath's energy is

$$U_{\text{approx}} = m_i c_i\,(T - T_{\text{ref}}) + m_w\,[L_f + c_w(T - T_{\text{ref}})] + \beta\, m_s. \tag{88}$$

Differentiating with the product rule, and using $\dot{m}_w = -\dot{m}_i$:

$$\dot{U}_{\text{approx}} = (m_i c_i + m_w c_w)\,\dot{T} + [(c_i - c_w)(T - T_{\text{ref}}) - L_f]\,\dot{m}_i + \beta\,\dot{m}_s. \tag{89}$$

Setting this equal to the heat flowing in, now including the drink's:

$$\begin{aligned} &H(T_{\text{room}} - T) + G(T_d - T) \\ &\quad = (m_i c_i + m_w c_w)\,\dot{T} - [L_f + (c_w - c_i)(T - T_{\text{ref}})]\,\dot{m}_i + \beta\,\dot{m}_s. \end{aligned} \tag{90}$$

At $T = T_{\text{ref}}$ this is the simpler balance (67). Elsewhere the heat to melt a kilogram of ice is $L_f + (c_w - c_i)(T - T_{\text{ref}})$, smaller below zero (Kirchhoff's law): at −15 °C, $333\,500 - 2080 \times 15 = 302\,300\,\text{J kg}^{-1}$, 9 % less. A more precise model would add the brine's salt-dependent heat capacity and density, non-ideal mixing, the measured NaCl liquidus, and the hydrated solid phase where it forms.

14.1 Consistent integrated energy at any starting and final temperature

Derivation. With an energy function, the integrated balance is exact: the heat let in equals the change of energy. The drink's heat flows inside the system and cancels (6), so for bath and drink together, with U_D from (60) and the undissolved salt's sensible heat included:

$$\boxed{\int_0^{t_f} H(T_{\text{room}} - T)\,\mathrm{d}t = \left[U_{\text{approx}} + (S_0 - m_s)\,c_s(T - T_{\text{ref}}) + U_D\right]_0^{t_f}} \tag{91}$$

Written out, with the bath at T_0 and the drink at $T_{d,0}$ at the start:

$$\begin{aligned} \int_0^{t_f} H(T_{\text{room}} - T)\,\mathrm{d}t = {} & m_{i,f} c_i (T_f - T_{\text{ref}}) - m_{i,0} c_i (T_0 - T_{\text{ref}}) \\ & + m_{w,f}\,[L_f + c_w(T_f - T_{\text{ref}})] - m_{w,0}\,[L_f + c_w(T_0 - T_{\text{ref}})] \\ & + \beta\,(m_{s,f} - m_{s,d,0}) + (S_0 - m_{s,f})\,c_s(T_f - T_{\text{ref}}) - S_0\,c_s(T_0 - T_{\text{ref}}) \\ & + C_d\,(T_{d,f} - T_{d,0}) - L_f\,m_{wd}\,\psi(T_{d,f}). \end{aligned} \tag{92}$$

This is the preferred endpoint equation for the idealised, constant-coefficient model. For a perfectly insulated box, set the left side to zero. For a real box over a finite time, the left side depends on the unknown history of T and cannot be found from the endpoints alone.

15 Model limits and practical next steps

1. **Initial temperature consistency.** The condition $T = -\alpha m_s/m_w$ cannot hold for pure ice and fresh water above 0 °C, or for brine that has not reached its ice yet. The first moments of a run are out of equilibrium; start the model once ice and brine have met, or add a non-equilibrium start.
2. **Salt concentration.** The $i_v K_f b$ law suits dilute brine only (Step 6). Use the measured NaCl freezing curve and eutectic for the bath's working range.
3. **Dissolution kinetics.** Rates depend on grain size, crust, and whether brine flows through the ice. Measure or fit k_m and $A_{s,0}$.
4. **The box.** $H = kA/d$ assumes flat, steady walls with faces at room and bath temperature. Calibrating H with a box test is more reliable than computing it.
5. **Switching of phases.** Enforce $m_i \geq 0$, $0 \leq m_s \leq S_0$ and saturation. Ice–brine equilibrium, saturated brine with salt, all-liquid and eutectic regimes each need their own closure.
6. **The can is not a cylinder.** Its neck and domed base change the volumes and areas of Step 9 by several per cent: measure the brimful volume and the headspace.
7. **Hold the can under.** A free can floats lying down with a third of its wall out of the brine, and cools 20 % more slowly (Table 3).
8. **The bath next to the can.** With no stirrer the brine near the can warms first, and the ice there melts first; one bath temperature overstates the cooling. The B8 report splits the bath into rings to follow this.
9. **The drink.** It stratifies slowly; its free convection weakens near 4 °C; it may supercool, then partly freeze at once; and a frozen drink may burst its can.
10. **Drink data.** The freezing points and heat capacities of Step 10 come from dilute laws and approximate partial heat capacities. Measure T_{fd} from a slow cooling curve (the plateau where ice appears) and c_d by mixing.
11. **Validation.** Compare predicted temperature–time curves and end states with experiments at several salt-to-water ratios, starting masses and temperatures, and insulation thicknesses; and, for the can, with probes in the drink and in the brine next to it, for full and part-filled cans, upright and lying, held and floating, and for drinks with and without ethanol.

16 Quick index of principal equations

First law with the can: (4)–(6). Conduction through the box: (7), with films (12). Ice and water conservation: (25). Latent energy: (19). NaCl energy: (28) and (29). Freezing depression: (34)–(36); the model's floor (38). Dissolution kinetics: (39)–(44). The drink and its can: (45), (46), (48). A part-filled can: (49)–(52); floating (53), (54); room to freeze (55). The ethanol model: (56)–(62). Dynamic derivatives: (71) and (74); the full system (76). Insulated equilibrium: (78)–(82). Finite-time energy: (83). True steady state: (87). Thermodynamically consistent update: (90)–(92).